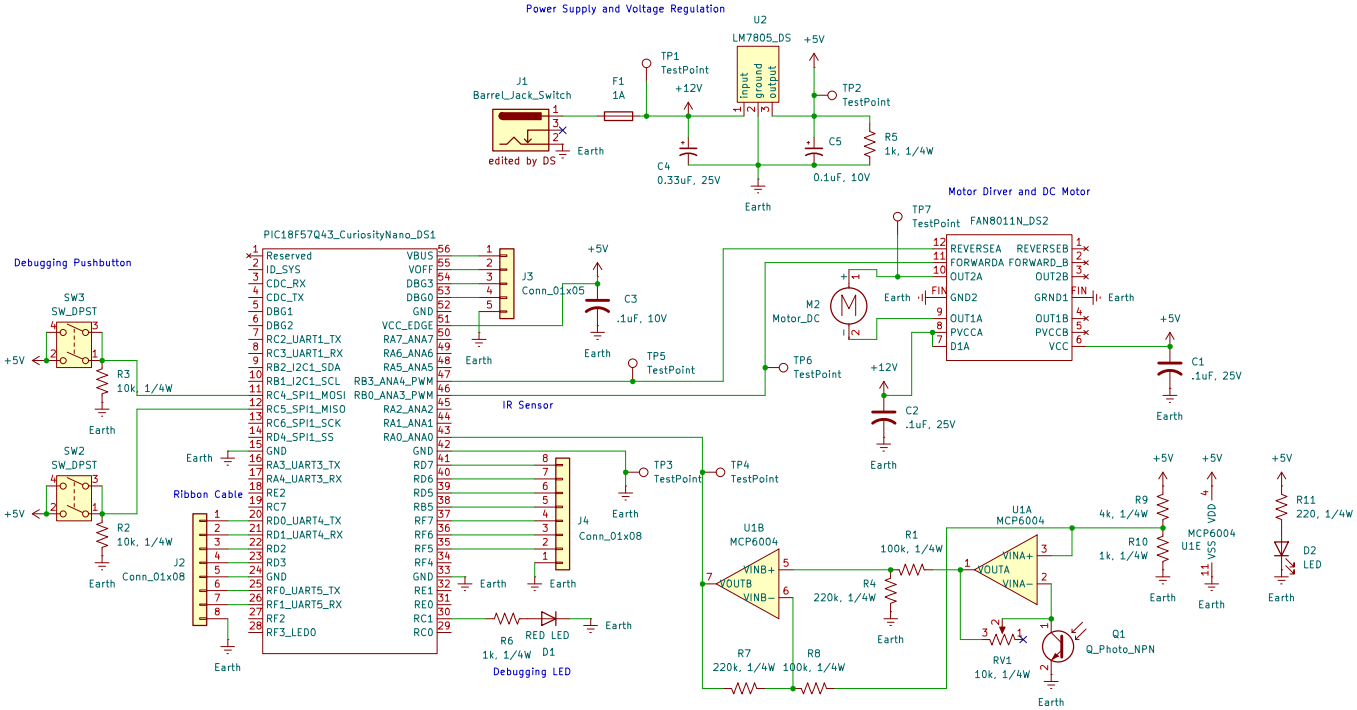
Power Supply and Voltage Regulation
U2
LM7805_DS
+5V
TP1
TestPoint
J1
Barrel_Jack_Switch
F1
1A
+12V
input
ground
output
TP2
TestPoint
edited by DS
Earth
C4
0.33uF, 25V
C5
0.1uF, 10V
R5
1k, 1/4W
Motor Dirver and DC Motor
TP7
TestPoint
FAN8011N_DS2
REVERSEA
FORWARDA
OUT2A
GND2
OUT1A
PVCCA
D1A
REVERSEB
FORWARD_B
OUT2B
GRND1
OUT1B
PVCCB
VCC
FIN
M2
Motor_DC
TP6
TestPoint
C1
.1uF, 25V
C2
.1uF, 25V
+12V
Debugging Pushbutton
SW3
SW_DPST
SW2
SW_DPST
R3
10k, 1/4W
R2
10k, 1/4W
PIC18F57Q43_CuriosityNano_DS1
Reserved
ID_SYS
CDC_RX
CDC_TX
DBG1
DBG2
RC2_UART1_TX
RC3_UART1_RX
RB2_I2C1_SDA
RB1_I2C1_SCL
RC4_SPI1_MOSI
RC5_SPI1_MISO
RC6_SPI1_SCK
RD4_SPI1_SS
GND
RA3_UART3_TX
RA4_UART3_RX
RE2
RC7
RD0_UART4_TX
RD1_UART4_RX
RD2
RD3
GND
RF0_UART5_TX
RF1_UART5_RX
RF2
RF3_LED0
VBUS
VOFF
DBG3
DBG0
GND
VCC_EDGE
RA7_ANA7
RA6_ANA6
RA5_ANA5
RB3_ANA4_PWM
RB0_ANA3_PWM
RA2_ANA2
RA1_ANA1
RA0_ANA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
J3
Conn_01x05
C3
.1uF, 10V
TP5
TestPoint
IR Sensor
TP3
TestPoint
TP4
TestPoint
Ribbon Cable
J2
Conn_01x08
J4
Conn_01x08
R6
1k, 1/4W
RED LED
D1
Debugging LED
U1B
MCP6004
VINB+
VOUTB
VINB−
R1
100k, 1/4W
R4
220k, 1/4W
R7
220k, 1/4W
R8
100k, 1/4W
U1A
MCP6004
VINA+
VOUTA
VINA−
RV1
10k, 1/4W
Q1
Q_Photo_NPN
R9
4k, 1/4W
R10
1k, 1/4W
U1E
MCP6004
VDD
VSS
R11
220, 1/4W
D2
LED